INCH-POUND

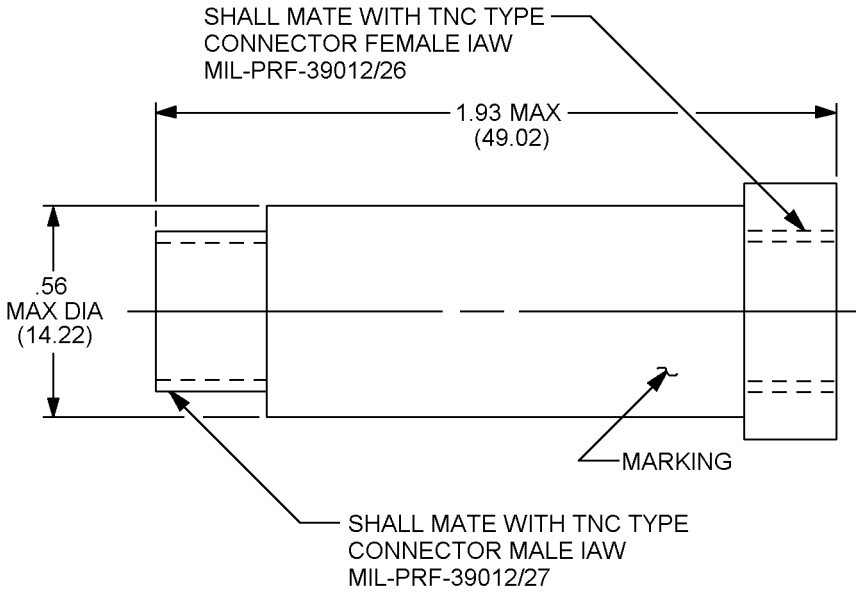
MIL-DTL-3933/17D
w/AMENDMENT 1
12 April 2021
SUPERSEDING
MIL-DTL-3933/17D
7 April 2011

DETAIL SPECIFICATION SHEET

ATTENUATORS, FIXED, SPACE LEVEL, NON SPACE LEVEL, COAXIAL LINE (TNC CONNECTORS), FREQUENCY RANGE: DC TO 18 GHz, CLASS IV, LOW POWER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-3933.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 1. Dimensions and configuration.

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ENGINEERING DATA:

Operating frequency range: DC to 4.5 GHz for PINs M3933/17-01 through -06; DC to 18 GHz for PINs M3933/17-07 through -24.

Test frequency ranges for SWR and attenuation measurements, measured or recorded, when using an Automatic Network Analyzer (ANA) or equivalent:

100 MHz (or lower) to 4.5 GHz for PINs M3933/17-01 through -06;
100 MHz (or lower) to 18 GHz for PINs M3933/17-07 through -24.

Basic test frequencies for sensitive measurements like connector repeatability, temperature sensitivity, power sensitivity and any other measurements where a 'delta' spec is imposed:

2.0 GHz for PINs M3933/17-01 through -06;
8.0 GHz for PINs M3933/17-07 through -24.

Operating temperature range: -55°C to 125°C.

Maximum weight: See table I.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Nominal impedance: 50 ohms.

Attenuation: See table I. Stability and sensitivity:

Maximum change in attenuation after:	Up to 10 dB, inclusive (dB)	Over 10 dB (dB/dB)
Temperature change or thermal shock	0.5	0.05
Vibration or shock	0.5	0.05
Moisture resistance or salt spray	0.5	0.05
Peak power	0.5	0.05

Frequency sensitivity, maximum: 0.4 dB/dB/GHz.
Temperature sensitivity of attenuation, maximum: 0.0006 dB/dB/°C.

Power: See table I.
Power sensitivity for full input power, maximum: 0.005 dB/dB/Watt.

VSWR: See table I.

Connector repeatability, maximum variation in attenuation: 0.04 dB.

Part or Identifying Number (PIN): M3933/17- (and dash number from table I with three device level designators are space level (T), screened (S) and non-screened (N) options).

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TABLE I. Electrical characteristics.

Dash number	Attenuation dB			Maximum power input at 25°C		VSWR maximum 1/					Weight (oz) Max	Dimension A (inches) maximum
	Nominal	Deviation				DC to 4.5 GHz	DC to 4.0 GHz	4.0 to 8.0 GHz	8.0 to 12.4 GHz	12.4 to 18.0 GHz		
		DC to 4.5 GHz	DC to 18 GHz	AV (W) 3/ (continuously)	Peak 2/ (kW)							
01 NST	1	±0.3	---	4	0.2	1.25:1	---	---	---	---	3.0	---
02 NST	2	±0.4	---	4	0.2	1.25:1	---	---	---	---	3.0	---
03 NST	3	±0.4	---	4	0.2	1.25:1	---	---	---	---	3.0	---
04 NST	4	±0.4	---	4	0.2	1.25:1	---	---	---	---	3.0	---
05 NST	5	±0.4	---	4	0.2	1.25:1	---	---	---	---	3.0	---
06 NST	6	±0.4	---	4	0.2	1.25:1	---	---	---	---	3.0	---
07 NST	20	---	±0.6	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
08 NST	30	---	±0.9	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
09 NST	1	---	±0.4	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
10 NST	2	---	±0.4	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
11 NST	3	---	±0.4	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
12 NST	4	---	±0.4	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---

See footnotes at end of table.

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TABLE I. Electrical characteristics - Continued.

Dash number		Attenuation dB			Maximum power input at 25°C		VSWR maximum 1/					Weight (oz) Max	Dimension A (inches) maximum
		Nominal	Deviation										
			DC to 4.5 GHz	DC to 18 GHz	AV (W) 3/ (continuously)	Peak 2/ (kW)	DC to 4.5 GHz	DC to 4.0 GHz	4.0 to 8.0 GHz	8.0 to 12.4 GHz	12.4 to 18.0 GHz		
13	NST	5	---	±0.4	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
14	NST	6	---	±0.4	2	0.2	---	1.15:1	1.20:1	1.15:1	1.35:1	3.0	---
15	NST	7	---	±0.5	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
16	NST	8	---	±0.5	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
17	NST	10	---	±0.5	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
18	NST	12	---	±0.5	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
19	NST	15	---	±0.5	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
20	NST	25	---	±0.8	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
21	NST	35	---	±0.9	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
22	NST	40	---	±1.0	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
23	NST	50	---	±1.4	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
24	NST	60	---	±1.5	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---

1/ VSWR value is for both ends.
2/ Peak power for a duty cycle of 5×10^{-4}, maximum pulse duration of 5 microseconds.
3/ Power input is derated linearly to 0.5 watts at +125°C.

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Referenced documents. In addition to MIL-DTL-3933, this specification sheet references
MIL-PRF-39012/26
MIL-PRF-39012/27

Amendment notations. The margins of this specification sheet are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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Navy – EC
Air Force - 85
DLA - CC

Review activities:
Army - MI
Navy - AS, MC, SH

Preparing activity:
DLA - CC

(Project 5985-2021-004)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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